Syllabus Details

Syllabus ID:	14161
Syllabus Name:	**Machine Learning in Data Science_Học máy trong khoa học dữ liệu**
Course Name English:	**Machine Learning in Data Science**
Subject Code:	**MDS301**
Learning-Teaching Method:	Inquiry-Based Learning,In-class lecture, Active Learning
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 104h self-study
Pre-Requisite:	PDS301m, MAE101
Description:	1. This course introduces the core concepts and practical techniques of machine learning for data science through practical, hands-on work with Python. 2. Students will learn how to preprocess and analyze data, implement common machine learning algorithms, and evaluate model performance using appropriate metrics. 3. Using modern tools such as Scikit-Learn and PyTorch, student will practice solving real data problems including prediction, classification, clustering and basic deep learning. 4. Additionally, students will gain basic experience deploying trained models through simple APIs and interpreting predictions with attention to ethical considerations.
StudentTasks:	- Students must attend at least 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises, assignments and labs given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the FU FLM at https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	Jupyter, Visual Studio Code, Git/GitHub, Docker (basic), FastAPI, Mlflow
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	377/QĐ-ĐHFPT dated 04/09/2026
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	4/9/2026

3 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Hands-On Machine Learning with Scikit-Learn and PyTorch: Concepts, Tools, and Techniques to Build Intelligent Systems	Aurélien Géron	O'Reilly Media	2025	1 st	979-8341607989	True	True	False	
2	Python Data Science Handbook: Essential Tools for Working with Data	VanderPlas, J.	O'Reilly Media	2023	2nd	978-1098121228	False	True	False	

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
3	Introduction to Machine Learning with Python: A Guide for Data Scientists	Müller, A. C., & Guido, S.	O'Reilly Media	2016	1st	978-1449369415	False	True	False	

7 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand Core Concepts: Explain fundamental machine learning principles of machine learning, distinguish between supervised, unsupervised, and reinforcement learning, and describe the end-to-end ML workflow.
2	CLO2	Prepare and Analyze Data: Proficiently preprocess datasets, perform exploratory data analysis, and engineer relevant features to support effective model training.
3	CLO3	Implement and Train Models: Implement and train a variety of key machine learning algorithms - including linear models, tree-based ensembles, and clustering methods - using standard Python machine learning libraries.
4	CLO4	Evaluate and Validate Models: Select and use appropriate metrics to evaluate model performance, diagnose issues such as overfitting or underfitting, and validate model performance using techniques like cross-validation.
5	CLO5	Apply Neural Networks: Understand the fundamentals of neural networks and build/train deep learning model using modern frameworks such as PyTorch for tasks including image classification.
6	CLO6	Deploy and Interpret Models: Demonstrate basic model deployment by exposing a trained model through a simple API and interpret model predictions while recognizing ethical and data bias considerations.
7	CLO7	Integrate ML solution into software system: Design, implement, and integrate a complete ML solution into a software system, including proper documentation, testing, and API deployment following software engineering best practices.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Summary of Math Prerequisites for Machine Learning: + Linear Algebra (vectors, matrices) + Probability & Statistics (probability distributions, mean, variance) + Differential Calculus (gradients, derivatives)	Offline	CLO1	T	Slides	MDS301	Read slides	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
2	Course introduction and Setup environment Introduction to Machine Learning (ML) + Definition of ML + Applications of ML + Types of ML (supervised, unsupervised, reinforcement learning)	Offline	CLO1	T	Slides, Textbook chapter 1, requirements.txt		Read textbook, slides Setup environment	
3	Introduction to Machine Learning (ML) + Main challenges of ML + Testing and Validating + ML vs Statistical Modeling	Offline	CLO1	T	Slides, Textbook chapter 1		Read textbook, slides Do exercises	
4	End-to-end ML workflow via small project + Problem framing and Performance Measure Selection + Data collection + Exploratory Data Analysis & Visualization + demo	Offline	CLO1, CLO2	TU	Slides, Textbook chapter 2	link4-1	Read textbook, slides Do project by following: https://github.com/ageron/handson-mlp	
5	End-to-end ML workflow via small project + Data preprocessing: Data Cleaning, Feature Engineering, Custom Transformers, Preprocessing pipelines + demo	Offline	CLO1, CLO2	TU	Slides, Textbook chapter 2	link5-1	Read textbook, slides Do project by following: https://github.com/ageron/handson-mlp	
6	End-to-end ML workflow via small project + Select and Train the model + Fine-Tune the model + demo	Offline	CLO1, CLO3	TU	Slides, Textbook chapter 2	link6-1	Read textbook, slides Do project by following: https://github.com/ageron/handson-mlp	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
7	Supervised learning: Introduction + Regression vs Classification + Typical supervised learning workflow + Training vs testing + Train/test split	Offline	CLO3, CLO4	T	Slides, Textbook chapter 3, Refference book		Read textbook, slides Do exercises	
8	Supervised Learning: Model Evaluation: + Metrics for regression + Metrics for classification + Overfitting and underfitting + demo	Offline	CLO3, CLO4	TU	Slides, Textbook chapter 3, Refference book		Read textbook, slides Do exercises	
9	Supervised Learning: Regression Algorithms + Linear Regression + Gradient Descent + demo	Offline	CLO3, CLO4	TU	Slides, Textbook chapter 4	link9-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
10	Supervised Learning: Regression Algorithms + Polynomial Regression + Learning Curves + demo	Offline	CLO3, CLO4	TU	Slides, Textbook chapter 4	link10-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
11	Supervised Learning: Regression Algorithms + Regularized Linear Models + Logistic Regression + demo	Offline	CLO3, CLO4	TU	Slides, Textbook chapter 4	link11-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
12	Supervised Learning: Classification Algorithms + K-nearest neighbors + Naïve Bayes Classifiers + demo	Offline	CLO3, CLO4	TU	Slides, 2nd Textbook chapter 2		Read textbook, slides Demo of K-nearest neighbors and naïve Bayes Classifiers in 2nd textbook chapter 2	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
13	Supervised Learning: Classification Algorithms + Decision Trees + Random Forests + demo	Offline	CLO3, CLO4	TU	Slides, 1st Textbook chap 5 and 6	link13-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
14	Supervised Learning: Classification Algorithms + Support Vertor Machines (SVM) + SVM Kernels + demo	Offline	CLO3, CLO4	TU	Slides, Introduction to Machine Learning with Python, chapter 2		Read textbook, slides Demo of SVM in Introduction to Machine Learning with Python, chapter 2	
15	Lab 01	Offline	CLO2, CLO3, CLO4	TU	Slides, Textbooks, Lab's questions		Do Lab	
16	Lab 01	Offline	CLO2, CLO3, CLO4	TU	Slides, Textbooks, Lab's questions		Do Lab	
17	Unsupervised Learning: Introduction + Introduction of Clustering and Dimensionality Reduction + Evaluation of Unsupervised Models: Internal vs. External metrics	Offline	CLO3, CLO4	T	Slides, Python Data Science Handbook, chapter 5; Introduction to Machine Learning with Python, chapter 3		Read textbook, slides Do exercises	
18	Unsupervised Learning: Clustering Algorithms + K-means + Density-Based Clustering (DBSCAN) + Demo	Offline	CLO3, CLO4	TU	Slides, Textbook chapter 8	link18-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
19	Unsupervised Learning: Clustering Algorithms + Hierarchical clustering: dendrogram, agglomerative clustering + Linkage criteria: single linkage, complete linkage, average linkage + Demo	Offline	CLO3, CLO4	TU	Slides, Textbook chapter 8	link19-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
20	Unsupervised Learning: Clustering Algorithms + Gaussian Mixtures + Demo	Offline	CLO3, CLO4	TU	Slides, Textbook chapter 8	link20-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
21	Unsupervised Learning: Dimensionality Reduction Algorithms + The curse of Dimensionality + PCA + Random Projection + LLE + Demo	Offline	CLO3, CLO4	TU	Slides, Textbook chapter 7	link21-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
22	Assignment 01	Offline	CLO1, CLO2, CLO3, CLO4	TU	Slides, Textbooks, Assignment's questions		Do Assignment at home	
23	Assignment 01	Offline	CLO1, CLO2, CLO3, CLO4	TU	Slides, Textbooks, Assignment's questions		Do Assignment at home	
24	Progress Test	Offline	CLO1, CLO2, CLO3, CLO4	U	Slides, Textbooks		Do test	
25	Basics of Neural Networks: + Artificial neuron + Perceptron + Activation Functions (ReLU, Sigmoid, Tanh) + Multilayer Perceptron (MLP) and Backpropagation + Optimizers (Adam, SGD) + demo	Offline	CLO5	TU	Slides, Textbook chapter 9 and 11	link25-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
26	Building and Training simple MLPs with Scikit-Learn: + Regression MLPs + demo	Offline	CLO5	TU	Slides, Textbook chapter 9 and 11	link26-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
27	Building and Training simple MLPs with Scikit-Learn: + Classification MLPs + demo Hyperparameter Tuning Guidelines: number of hidden layers, number of neurons per hidden layer, learning rate, batch size	Offline	CLO5	TU	Slides, Textbook chapter 9 and 11	link27-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
28	Introduction of Deep Learning Frameworks: + What is framework? Why we need frameworks? + Simple comparison of TensorFlow/Keras and PyTorch + PyTorch fundamentals: tensor, hardware acceleration, autograd + Building an Image Classifier with PyTorch (demo)	Offline	CLO5	TU	Slides, Textbook chapter 10	link28-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
29	Deep learning for image with PyTorch: + Convolutional Neural Networks (CNN): convolutional layers, pooling layers, CNN architectures	Offline	CLO5	T	Slides, Textbook chapter 12		Read textbook, slides Do exercises	
30	Deep learning for image with PyTorch: + Implementing convolutional layers, pooling layers, ResNet-34 in PyTorch + demo	Offline	CLO5	TU	Slides, Textbook chapter 12	link30-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
31	Transfer Learning: + Pretrained models for transfer learning (TorchVision) + Fine-tuning pretrained models + Regularization + Demo	Offline	CLO5	TU	Slides, Textbook chapter 12	link31-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
32	Lab 02	Offline	CLO5	TU	Slides, Textbooks, Lab's questions		Do Lab	
33	Lab 02	Offline	CLO5	TU	Slides, Textbooks, Lab's questions		Do Lab	
34	MLOps introduction Model deployment: + Definition of Model deployment + Deployment workflow + Model serialization: save and load trained model (with demo)	Offline	CLO6	TU	Slides		Read slides Demo by following teacher	
35	Model deployment: + Build prediction model + Deploy model as REST API (Flask/FastAPI) + demo	Offline	CLO6	TU	Slides		Read slides Demo by following teacher	
36	Model deployment: + Containerization: concept and workflow, tools presentation (Docker, Kubernetes) + Cloud deployment: concept, architecture, cloud service providers (AWS, Azure, GCP)	Offline	CLO6	TU	Slides		Read slides, self learning for coding deployment	
37	Model Monitoring: + Definition of model monitoring + Drift Detection: Data drift, concept drift	Offline	CLO6	TU	Slides		Read slides, Do exercises	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
38	Model Monitoring: + Performance decay + Prediction distribution monitoring + Logging predictions	Offline	CLO6	TU	Slides		Read slides, Do exercises	
39	Model Monitoring: + Simple monitoring demo + Practical monitoring workflow	Offline	CLO6	TU	Slides		Read slides Demo by following teacher	
40	ML Pipelines and Software Engineering practices for ML Systems + ML Pipelines: - Pipeline concepts - Scikit-learn Pipelines (build pipeline, train, predict, evaluate) - demo + Software Engineering practices for ML Systems: - Version control for ML projects (Git workflow) - Repository structure for ML projects - Unit testing for ML components - Introduction to CI/CD for ML pipelines	Offline	CLO6, CLO7	TU	Slides		Read slides Demo by following teacher	
41	MLFlow and Experiment Tracking: + Experiment tracking + Model versioning + Reproducibility + Demo	Offline	CLO6, CLO7	TU	Slides	link41-1	Read slides Demo by following teacher	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
42	Assignment 02 guides: Assignment 02 Requirements: + ML pipeline implementation + Model deployment as REST API + GitHub repository submission + Documentation (README)	Offline	CLO5, CLO6	TU	Slides, Textbook,		Do Assignment and commit to Github	
43	Assignment 02 review	Offline	CLO5, CLO6	TU	Slides, Textbook,		Do Assignment and commit to Github	
44	Assignment 02 review	Offline	CLO5, CLO6	TU	Slides, Textbook,		Do Assignment and commit to Github	
45	Time Series Forecasting: + Time Series concepts + ARMA model family + demo	Offline	CLO6	TU	Slides, Textbook chapter 13	link45-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
46	Time Series Forecasting: + Prophet (Meta)	Offline	CLO6	TU	Slides		Read textbook, slides	
47	Time Series Forecasting: + Basics of Recurrent Neural Networks (RNNs) + LSTM model + demo	Offline	CLO6	TU	Slides, Textbook chapter 13	link47-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
48	Natural Language Processing (NLP): + NLP basics and workflow + Tokenization, Bag-of-Words, TF-IDF + demo	Offline	CLO6	TU	Slides, Textbook chapter 14 and chapter 15	link48-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
49	Natural Language Processing (NLP): + Word Embeddings (Word2Vec) + demo	Offline	CLO6	TU	Slides, Textbook chapter 14 and chapter 15	link49-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
50	Natural Language Processing (NLP): + Transformers & BERT using Hugging Face + demo: sentiment analysis with Hugging Face	Offline	CLO6	TU	Slides, Textbook chapter 14 and chapter 15	link50-1	Read textbook, slides Demo by following: https://github.com/ageron/handson-mlp	
51	Recommendation Systems: + Recommendation system + Content-Based Filtering + demo	Offline	CLO6	TU	Slides		Read slides Demo by following teacher	
52	Recommendation Systems: + Collaborative Filtering + demo	Offline	CLO6	TU	Slides		Read slides Demo by following teacher	
53	Recommendation Systems: + Matrix Factorization + demo	Offline	CLO6	TU	Slides		Read slides Demo by following teacher	
54	AI Ethics and Responsible AI: + Responsible AI principles + Ethical risks in AI system	Offline	CLO6	TU	Slides		Read slides	
55	Final Project Presentation and Demo: + system architecture + ML model + deployed API demo + Q&A	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7	U			Final reports + demo	
56	Final Project Presentation and Demo: + system architecture + ML model + deployed API demo + Q&A	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7	U			Final reports + demo	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
57	Final Project Presentation and Demo: + system architecture + ML model + deployed API demo + Q&A	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7	U			Final reports + demo	
58	Final Project Presentation and Demo: + system architecture + ML model + deployed API demo + Q&A	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7	U			Final reports + demo	
59	Final Project Presentation and Demo: + system architecture + ML model + deployed API demo + Q&A	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7	U			Final reports + demo	
60	Final Project Presentation and Demo: + system architecture + ML model + deployed API demo + Q&A	Offline	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7	U			Final reports + demo	

0 Constructive question(s)

5 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	2	20.0%	>0	At home	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	Homework, mini-projects, and optionally the constructivist approach as proposed by the lecturer.	by instructor	All subjects in syllabus	in class, by instructor	

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
2	Lab	on-going	2	20.0%	>0	90	CLO2, CLO3, CLO4, CLO5, CLO6	Practical Lab	by instructor		Individual or team work, guided by instructor, submission by a given deadline	Instructor has resposibility to guide and review the lab practice for students.
3	Progress Test	on-going	1	10.0%	>0	30'	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	Multiple choices Marked by Computer	25		in class, by instructor	
4	Project	on-going	1	20.0%	4	At home	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7	Homework, mini-projects, and optionally the constructivist approach as proposed by the lecturer.	by instructor	All Modules	in class, by instructor	Instructor has resposibility to review for students. Submit Exercise 1 on slot 13
5	Final exam	Final exam	1	30.0%	4	60'	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7	Multiple choices Marked by Computer	50	All Modules	by exam board, using computer	